Global Renewable Energy in 2025: How Far Have We Come?

A data-driven overview of renewables across primary energy, electricity, hydropower, wind, solar, and regional transition gaps.

~14%

Primary energy from renewables

2024, substitution method

~30%

Electricity from renewables

2025 global average

~9,600 TWh

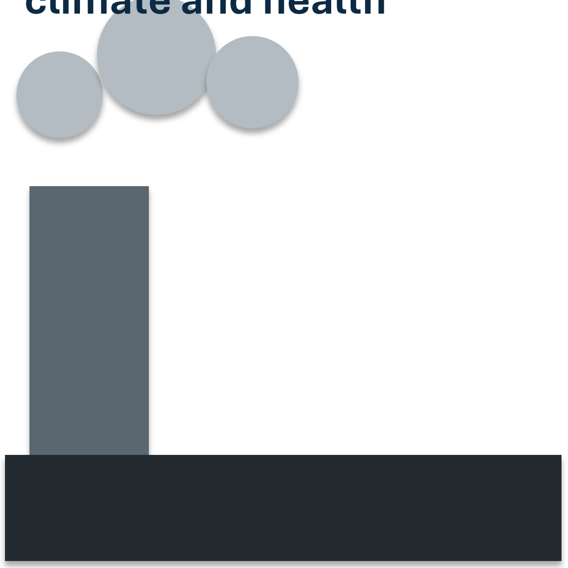
Renewable electricity output

Brief technology total

Sources: Our World in Data; Ember (2026); Energy Institute Statistical Review of World Energy (2025)

75% of Global GHG Emissions Still Come from Fossil Fuels

The fossil fuel problem is both climate and health



Energy systems remain shaped by fossil-fuel dominance since the Industrial Revolution.

75%

of global greenhouse gas emissions come from burning fossil fuels for energy

5M+

premature deaths per year from local air pollution linked to fossil fuel combustion

Rapid shift

needed toward low-carbon energy: renewables and nuclear

Core implication: renewables are not only a power-sector story; they are central to emissions and air-quality strategy.

Scope of modern renewables: solar, wind, hydropower, geothermal, bioenergy, wave and tidal; traditional biomass is typically excluded.

Policy lens: deployment pace must address climate risk while lowering fossil combustion pollution exposure.

The transition challenge: displace fossil fuels across electricity, transport, and heating—not just add clean supply.

Sources: Our World in Data; Ember (2026); Energy Institute Statistical Review (2025)

PRIMARY ENERGY MIX

Renewables Now Supply ~14% of Global Primary Energy

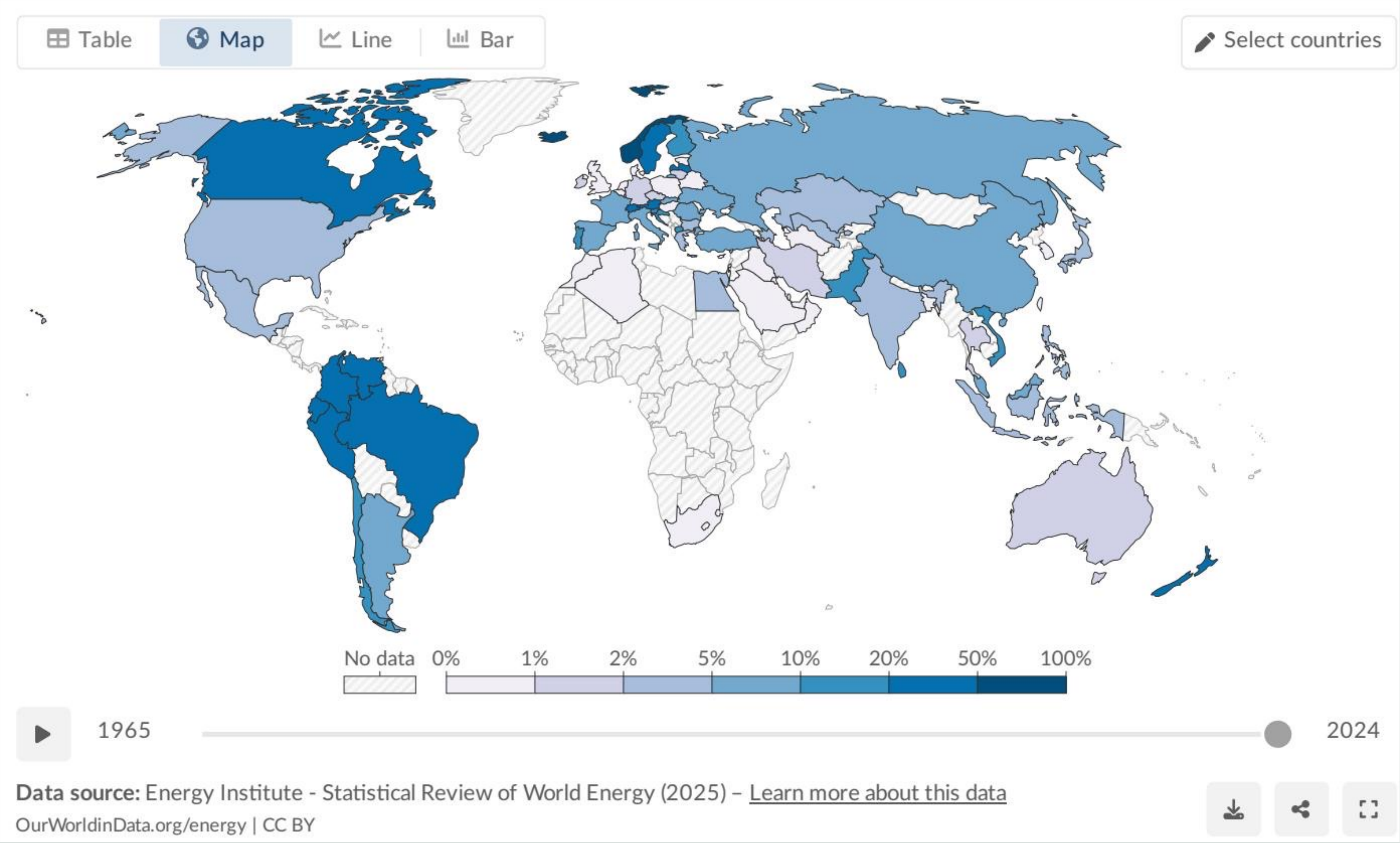


Figure: share of primary energy from renewables by country, 2024.

~14%

of global primary energy comes from renewables

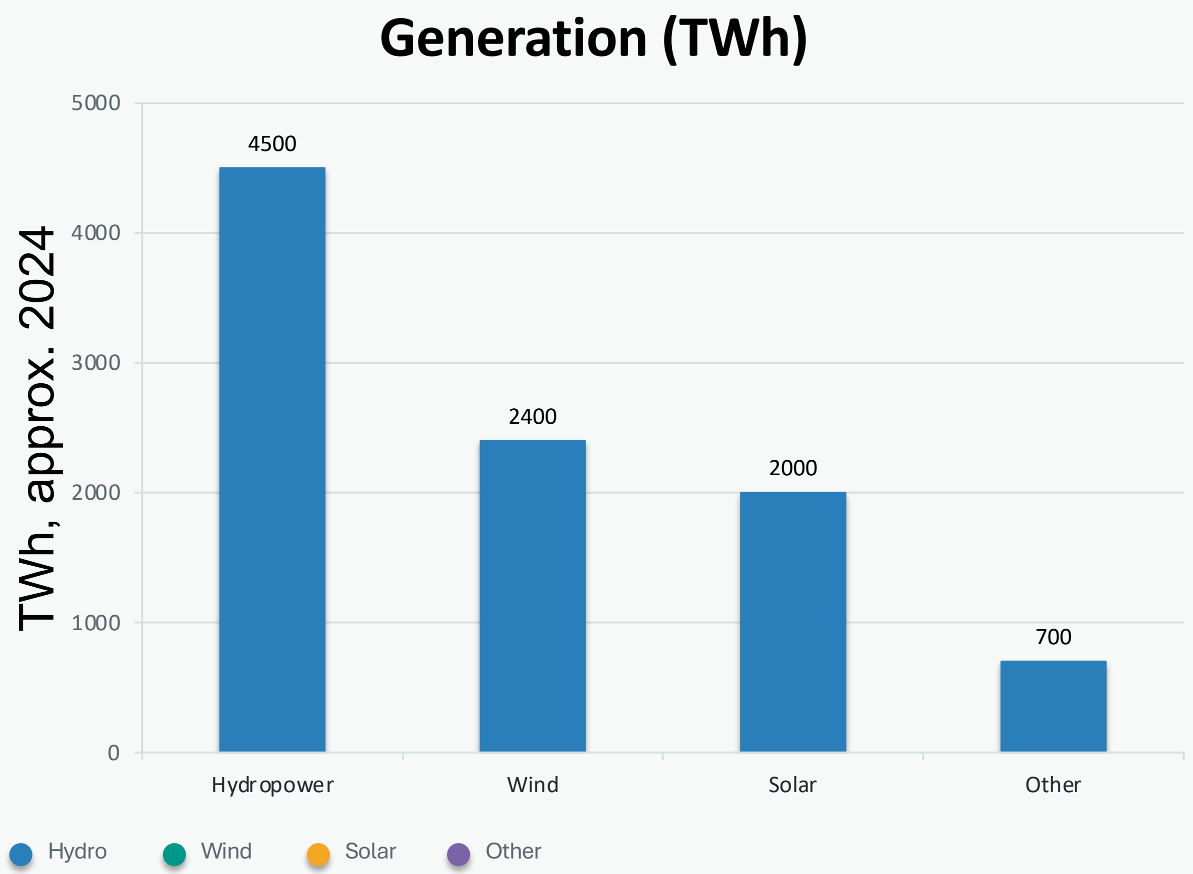
2024 data, substitution method

Why lower than electricity: primary energy includes transport and heating, which remain more reliant on oil and gas.

Leaders: countries such as Norway, Brazil and New Zealand derive large shares from renewables.

Laggards: fossil-fuel-producing nations in the Middle East and parts of Asia remain below 5%.

Renewable Electricity Generation Is Growing Rapidly, Led by Wind and Solar



Technology	Generation	Share
Hydropower	~4,500 TWh	~47%
Wind	~2,400 TWh	~25%
Solar	~2,000 TWh	~21%
Other renewables	~700 TWh	~7%

Hydro still leads: roughly half of renewable electricity generation.

Wind + solar: combined output now nearly matches hydropower.

Growth signal: wind and solar have accelerated strongly since around 2010.

Technology generation values are approximate from the source brief.

Almost One-Third of Global Electricity Now Comes from Renewables

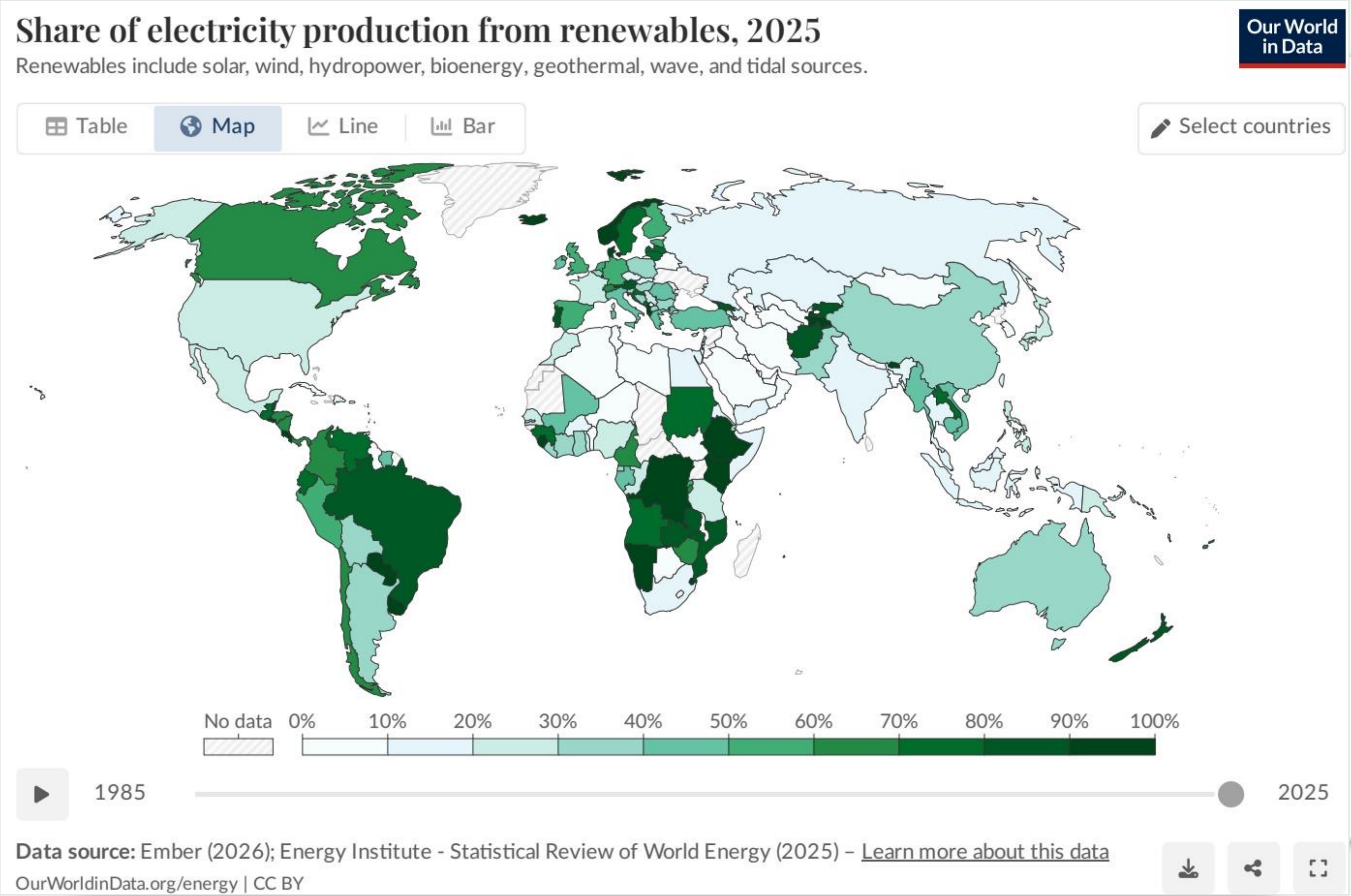


Figure: share of electricity from renewables by country, 2025.

~30%

World average

60–90%+

leaders such as Brazil, Canada, Nordics

Country / region	Renewable electricity share
Norway	~98%
Brazil	~85%
Canada	~65%
China	~35%
India	~25%
Saudi Arabia	<2%

Electricity is where renewables have made the most progress; transport and heating keep total primary energy lower.

Hydropower Remains the Largest Renewable Source at ~50% of Generation

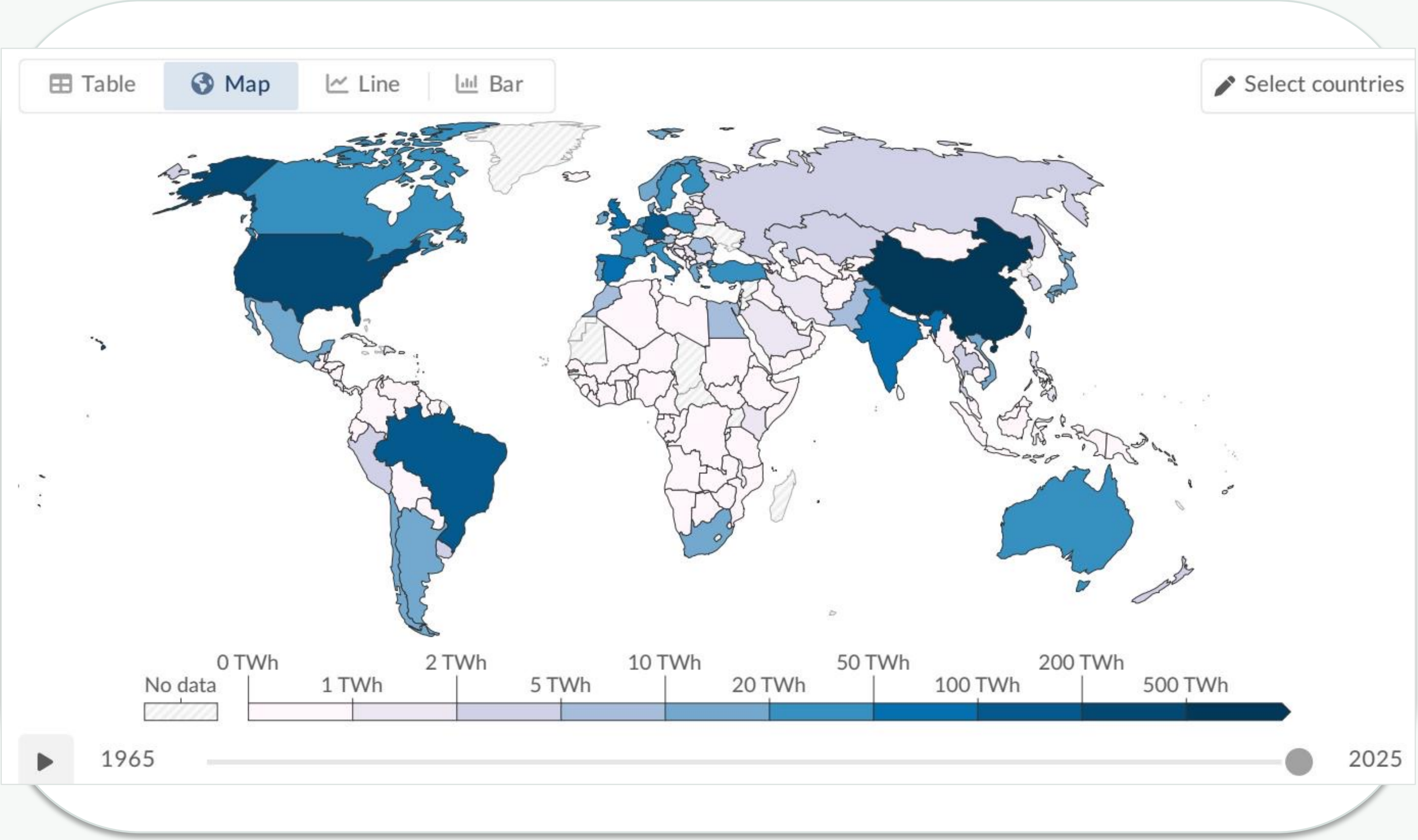


Figure: hydropower generation by country, 2025.

~4,500 TWh

global hydropower generation

approx. 2024 from brief

~47–50%

of renewable electricity generation

Top producers: China, Brazil and Canada.

Strategic role: hydropower has generated electricity at scale for more than a century.

Constraint: future growth potential is more limited than wind and solar.

Sources: Our World in Data; Ember (2026); Energy Institute Statistical Review (2025)

Wind and Solar Are the Fastest-Growing Energy Technologies in History

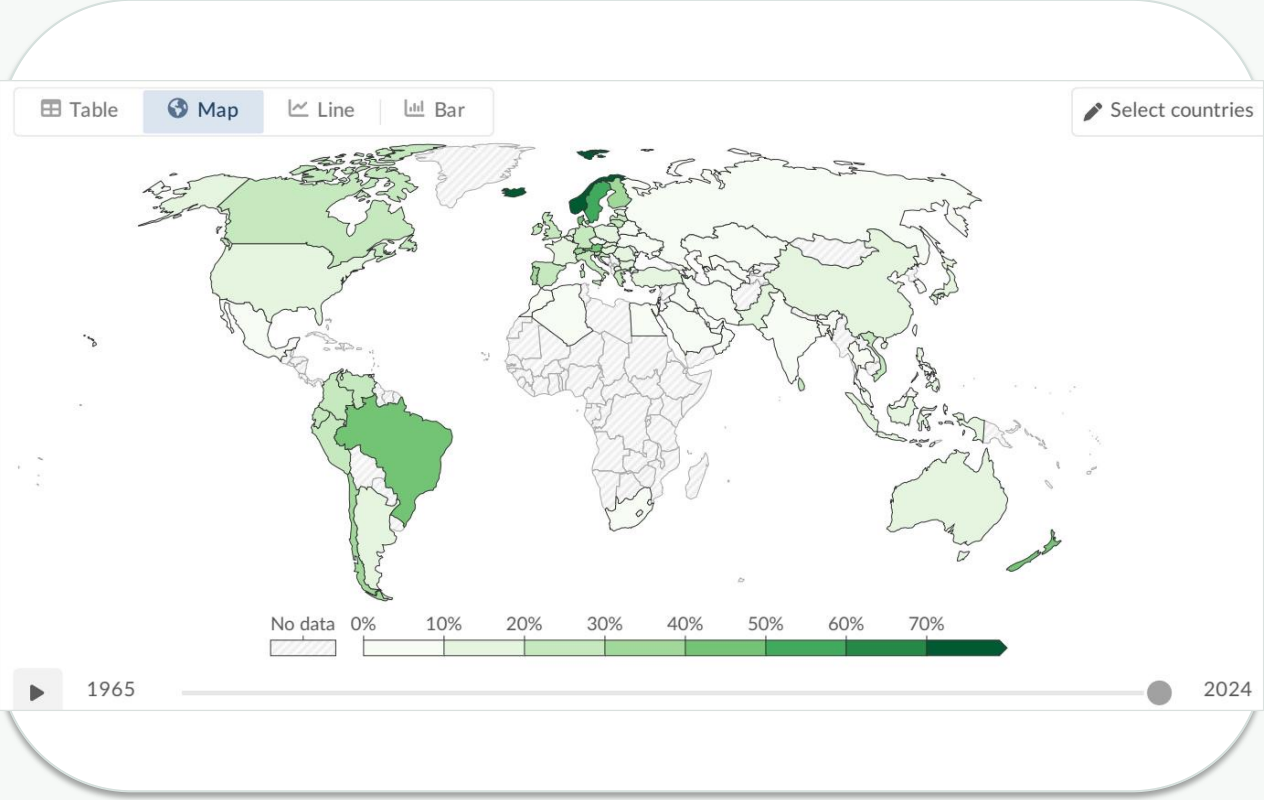
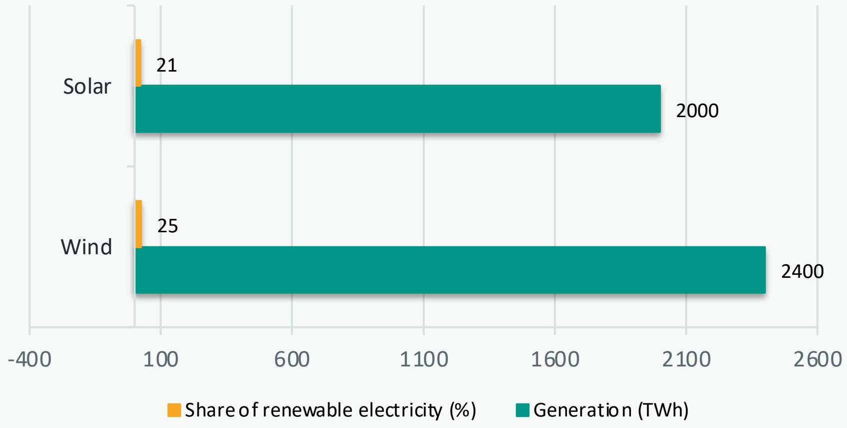


Figure: wind power generation by country, 2025.



~2,400 TWh

Wind: ~25% of renewable electricity

~2,000 TWh

Solar: ~21% of renewable electricity

Signal for strategy: solar is the fastest-growing source, while wind is already a major contributor in Europe, China and the United States; together they nearly match hydropower's output.

Stark Regional Gaps: Sub-Saharan Africa Lags While Scandinavia and Latin America

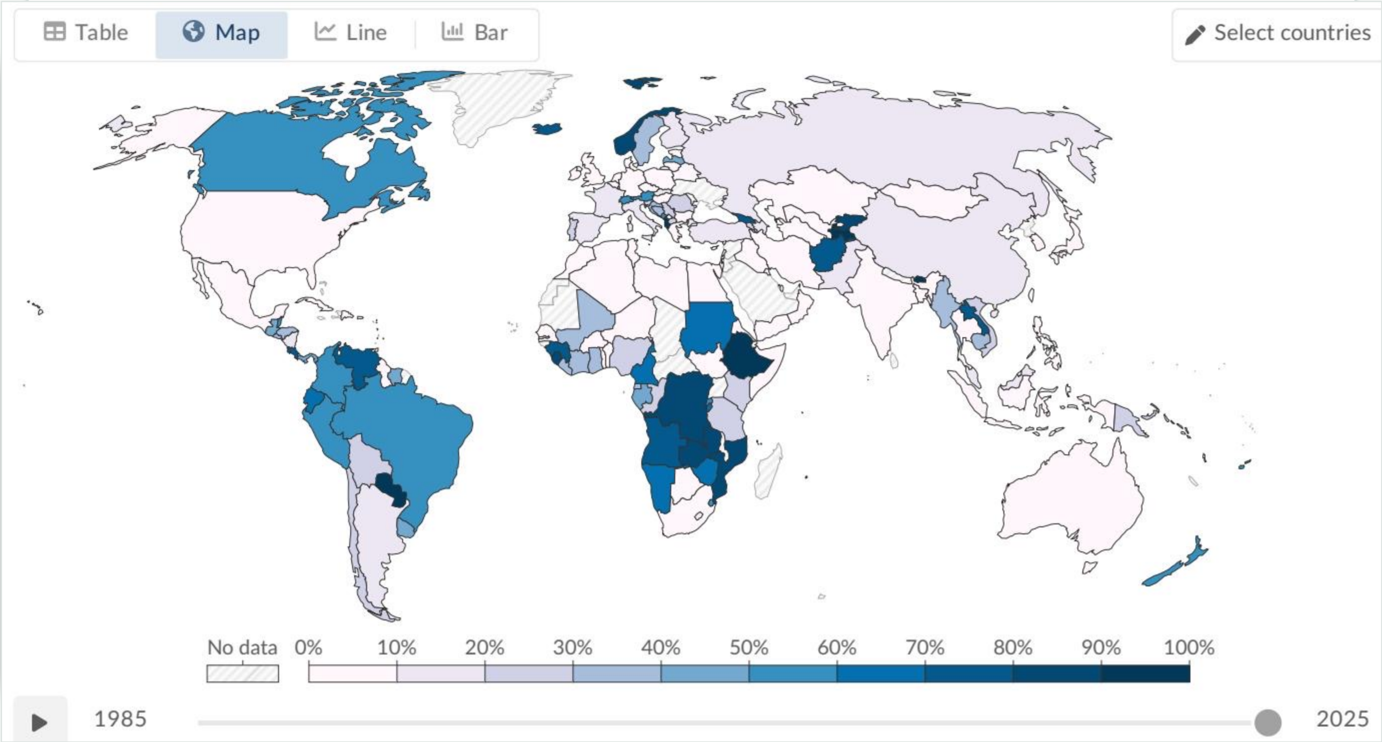


Figure: share of electricity from hydropower by country, 2025.

Uneven transition drivers

Hydro endowment

Scandinavia, Brazil and Canada benefit from abundant hydropower resources.

Policy investment

Denmark and Germany have invested heavily in wind and solar.

Infrastructure gap

Much of Sub-Saharan Africa has minimal renewable electricity infrastructure.

Fossil dependence

Many Middle Eastern nations remain almost entirely fossil-fuel dependent for electricity.

The same global average masks systems ranging from ~98% renewable electricity in Norway to <2% in Saudi Arabia.

Sources: Our World in Data; Ember (2026); Energy Institute Statistical Review (2025)

Key Takeaways: Renewables Are Growing Fast but the Transition Must Accelerate

Scale reached

~14% of primary energy and ~30% of electricity now come from renewables.

Technology mix

Hydropower remains dominant, but wind and solar combined nearly match hydro output.

Regional divide

Scandinavia, Latin America and Canada lead; much of the Middle East and Sub-Saharan Africa remains below 10%.

Sector gap

Transport and heating lag behind electricity, keeping total primary energy lower.

Forward-looking call: accelerate clean electricity deployment while extending decarbonization into transport and heating.

Sources: Our World in Data; Ember (2026); Energy Institute Statistical Review (2025)